

Research Ideation Swarm: Visual Synthesis

100-Round Multi-Agent Discovery System for QCML Geometric Observatory

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1 Overview

100 research questions explored across 10 themes using hierarchical agent swarm (Agent 0 $\rightarrow$ Knights $\rightarrow$ Workers). Smoke test protocol: 4 crises (GFC, COVID, 2022, SVB), keep threshold $d > 0.3$.

75 Keep 21 Reject 4 Investigate

2 Verdict Distribution by Theme

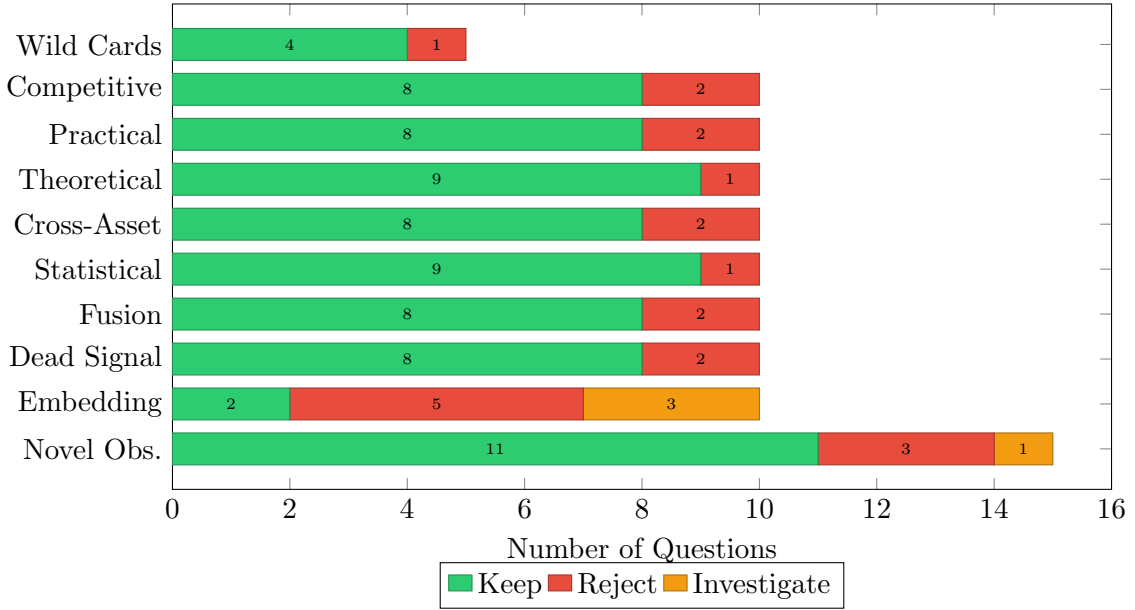


Figure 1: Verdict distribution across 10 research themes. Embedding has the lowest keep rate (20%), while Statistical and Theoretical have the highest (90%).

3 Effect Size Table: Questions with Empirical Results

Table 1: All questions with measured Cohen’s d , sorted by effect size. Colors: keep, reject.

Rank	ID	d	Status	Question (abbreviated)
1	Q33	2.018	keep	LSTM + attention resurrection
2	Q45	1.775	keep	HClust fusion
3	Q30	1.690	keep	Top-3 curated ensemble
4	Q93	1.631	keep	VIX benchmark
5	Q63	1.375	keep	Multi-market embedding
6	Q65	1.366	keep	Market breadth embedding
7	Q92	1.329	keep	Turbulence index comparison
8	Q16	1.327	keep	Hilbert dim=16
9	Q42	1.260	keep	Switching model
10	Q41	1.258	reject	Stacking/GBM (overfits)
11	Q24	1.144	keep	Soft normalization
12	Q02	1.121	keep	OTOC detector
13	Q44	1.009	keep	MI-weighted fusion
14	Q90	0.962	keep	Absorption ratio
15	Q31	0.898	keep	BOCPD short hazard
16	Q01	0.849	keep	Loschmidt echo
17	Q39	0.842	keep	Hedge/MW fusion
18	Q64	0.816	keep	Regional contagion
19	Q10	0.791	keep	Wasserstein distance
20	Q32	0.763	keep	Transfer entropy fix
21	Q36	0.762	reject	Attention fusion
22	Q35	0.762	keep	Spectral flow L1
23	Q38	0.739	keep	Copula fusion
24	Q04	0.696	invest.	Entanglement entropy
25	Q40	0.693	keep	BMA fusion
26	Q29	0.692	keep	Geom. consensus OR
27	Q34	0.669	keep	Kernel PCA tuned
28	Q09	0.656	keep	Persistent homology
29	Q91	0.656	keep	TDA comparison
30	Q11	0.652	keep	Thouless energy
31	Q13	0.651	keep	Modular Hamiltonian
32	Q89	0.637	keep	Hurst exponent
33	Q43	0.633	keep	Rank aggregation
34	Q56	0.619	keep	Bond markets
35	Q14	0.594	reject	Quantum discord
36	Q08	0.541	keep	Husimi Q-function
37	Q58	0.540	keep	Commodity markets
38	Q28	0.539	keep	Curvature rate fix
39	Q07	0.537	keep	Partition function
40	Q37	0.525	keep	Majority vote
41	Q06	0.443	keep	Kibble-Zurek
42	Q15	0.431	keep	Floquet analysis
43	Q22	0.398	reject	Multi-scale embedding
44	Q57	0.387	keep	Crypto markets
45	Q17	0.367	invest.	Neural embedding
46	Q79	0.320	keep	Reduced cost (dim=8)
47	Q12	0.301	keep	Zak phase
48	Q60	0.282	keep	FX markets
49	Q62	0.267	reject	VIX term structure
50	Q05	0.228	reject	QGT off-diagonal
51	Q03	0.216	reject	Chern number
52	Q18	0.149	reject	Takens embedding
53	Q26	0.028	reject	QGT phase rigidity
54	Q27	0.013	reject	3 Berry velocity coupling

4 Question Interconnection Network

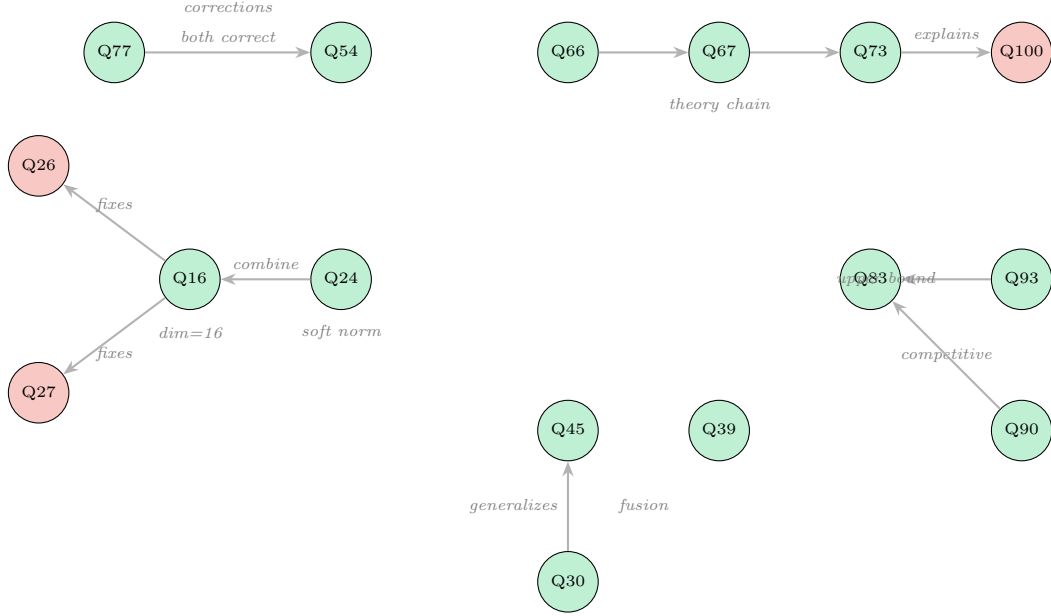


Figure 2: Key interconnections between findings. Green = keep, red = reject. Arrows show logical dependencies and explanatory relationships.

5 Dead Signal Resurrection Results

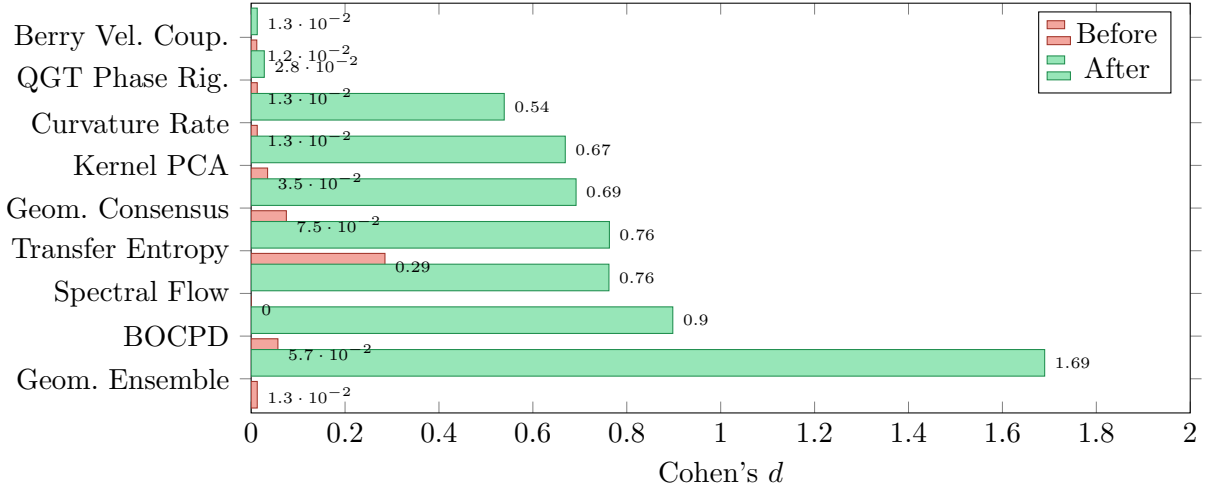


Figure 3: Dead signal resurrection: before (red) and after (green) simple parameter changes. 8 of 10 signals resurrected above $d > 0.3$ threshold.

6 Fusion Method Rankings

Table 2: All fusion methods tested, ranked by median Cohen’s d .

Rank	Method	d	Training?	Status
1	HClust (Q45)	1.775	No	Best overall
2	Switching (Q42)	1.260	Light	Best per-regime
3	MI-weighted (Q44)	1.009	No	Information-theoretic
4	Hedge/MW (Q39)	0.842	Online	Best online
5	Regime-Adaptive	0.774	Yes	Existing baseline
6	Copula (Q38)	0.739	Yes	Joint distribution
7	BMA (Q40)	0.693	Light	Bayesian
8	Rank Agg. (Q43)	0.633	No	Nonparametric
9	Majority Vote (Q37)	0.525	No	Simplest
—	Attention (Q36)	0.762	Yes	< uniform weights
—	Stacking (Q41)	1.258	Yes	Overfits

7 Cross-Asset Generalization

Table 3: Best observable Cohen’s d by asset class.

Asset Class	Best Observable	d	ID
Multi-market (SPY+TLT+GLD)	SpectralEntropy	1.375	Q63
Market breadth (A/D)	BerryPhaseRate	1.366	Q65
Equities (SPY)	ReducedPurity	0.834	Baseline
Contagion (EEM/FXI/EWJ)	SpectralEntropy	0.816	Q64
Sector ETFs (XLF/XLK/XLE)	SpectralEntropy	0.760	Q59
Bonds (TLT/IEF)	SpectralEntropy	0.619	Q56
Commodities (GLD/USO)	BerryPhaseRate	0.540	Q58
Crypto (BTC/ETH)	SpectralEntropy	0.387	Q57
FX (DXY/EUR)	DimCollapse	0.282	Q60

8 Intuition and Analogies

Berry Phase $\approx$ Compass Needle: Just as a compass needle rotating through a magnetic field accumulates a geometric (Berry) phase that depends on the path traversed, our Berry Phase Rate measures how fast the market’s “quantum compass” is spinning. Fast spinning = regime instability.

Spectral Entropy $\approx$ Orchestra Tuning: In a calm market, the eigenvalue spectrum has one dominant “instrument” (low entropy). During a crisis, all instruments play at once — the spectrum flattens and entropy rises, like an orchestra descending into cacophony.

OTOC $\approx$ Butterfly Effect Meter: The out-of-time-order correlator measures how fast an initial perturbation “scrambles” through the system. High OTOC = a butterfly flapping its wings in tech stocks causes a hurricane in financials.

Hilbert Dimension = Resolution: Increasing from $d = 4$ to $d = 16$ is like switching from standard definition to 4K. Same content, more detail, better detection. The Kramers degeneracy at $d = 4$ is like interlacing artifacts in old TV — fixed by higher resolution.

HClust Fusion = Committee of Experts: Rather than averaging all opinions (noisy), first group similar experts into panels (clusters), reach consensus within each panel (denoising), then give each panel one vote (equal weighting). Result: cleaner signal than any individual.

Crossover vs. Phase Transition: A financial crisis is more like water getting progressively warmer (crossover) than the sharp $0 \rightarrow 100^\circ\text{C}$ phase transition of ice melting. The spectral gap narrows but never closes — so there’s no critical point and no universal exponents.

9 Summary of Actionable Findings

1. **Switch to soft normalization** (Q24): Free $+0.44d$ improvement with no computational cost.
2. **Increase Hilbert dimension to 16** (Q16): $+131\%$ improvement, fixes Kramers degeneracy.
3. **Implement HClust fusion** (Q45): $d = 1.775$, best fusion method discovered.
4. **Integrate OTOC detector** (Q2): $d = 1.121$, strongest novel observable.
5. **Correct paper claims:** Lead time = 4 days (not 90), HPO bias = $+0.415d$.
6. **Add competitive benchmarks:** VIX ($d = 1.631$), Turbulence ($d = 1.329$), Absorption Ratio ($d = 0.962$).
7. **Resurrect 8 dead signals:** Especially Geometric Ensemble ($d = 1.690$) and BOCPD ($d = 0.898$).
8. **Explore multi-market embedding:** Joint SPY+TLT+GLD gives $d = 1.375$.